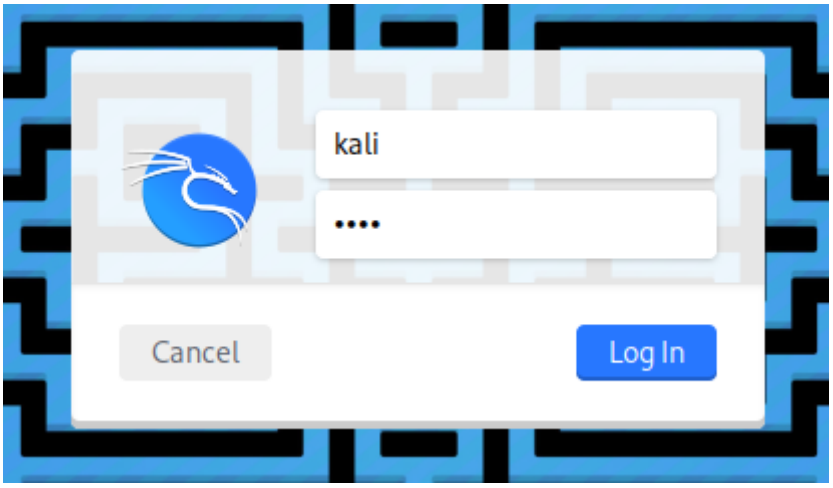
GERMAN-MALAYSIAN INSTITUTE
Training for Advanced Technology

COMPUTER & INFORMATION DEPARTMENT

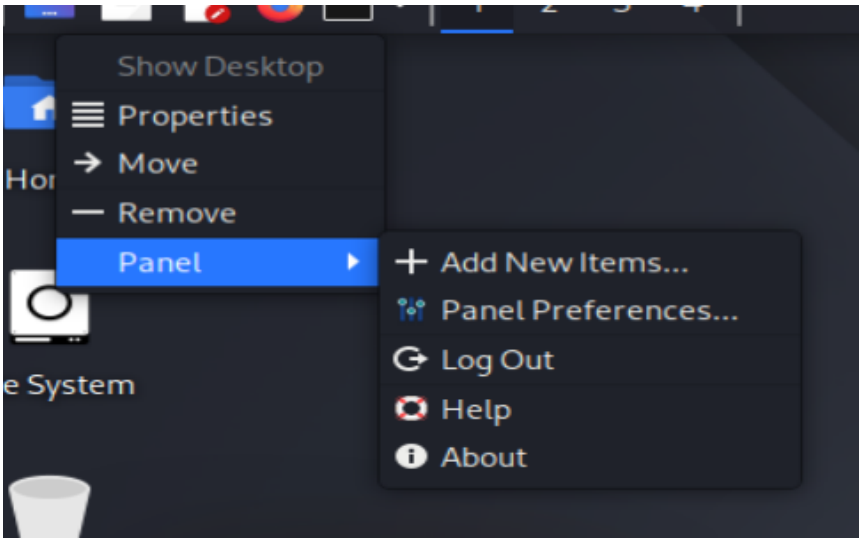
CBS 2353 - Penetration Testing

1.3.7 Lab - Investigate Kali Linux

Student's Name	**ABDUL HARIZ B. MOHD SARNUDIN**
ID Number	**CBS24070208**
Trade/Group/Sem	**DCBS / CBS 1 / SEMESTER 4**
TTO's Name	**Ts Noormelah Shamsul**



Logon with credentials = kali:kali



Optional, user can add or edit the panels



Application menu

1.0

Step 1: Command documentation

a. To learn more about the man page, open a terminal, and type:

```
┌──(kali㉿Kali)-[~]
└─$ man man
```

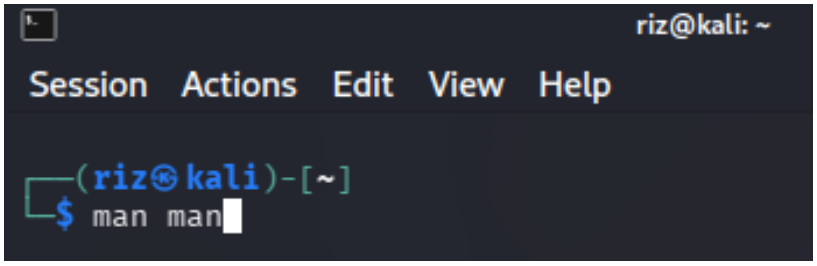
Scroll through the output and name a few sections that are included in a man page.

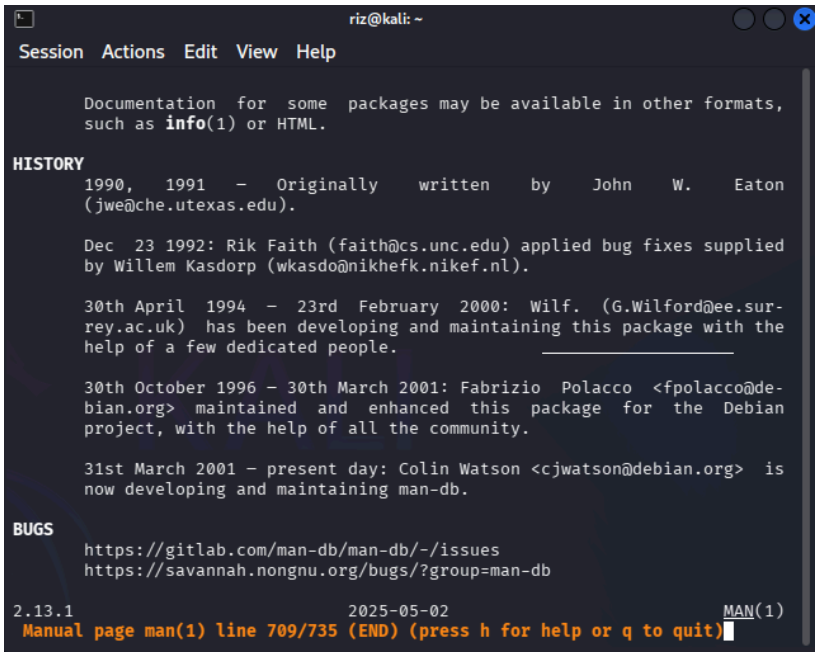
Answer Area

Enter your answer here

Show Answer

Answer :





2.0

Step 2: Create and change directories.

In this step, you will use the change directory (**cd**), make directory (**mkdir**), and list directory (**ls**) commands.

Note: A directory is another word for folder. The terms directory and folder are used interchangeably throughout this lab.

Note: There may be times when a command will not work because the user that is currently logged on does not have permission to perform it. To temporarily gain permission for the command, proceed the command with **sudo**, which stands for super user "do". You may need to provide the password of a user that has permission to perform the command. Alternatively, you can use a terminal that has a higher permission level. This terminal is called Root Terminal Emulator and can be found in the panel by using the drop-down menu next to the terminal icon.

a. In a terminal, enter **pwd** at the prompt. This command will print the current working directory to the terminal.

```
┌──(kali㉿Kali)-[~]
└─$ pwd
/home/kali
```

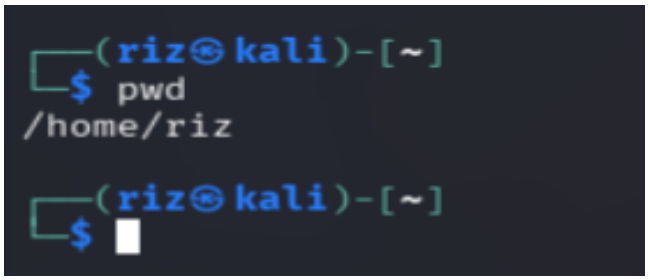
What is the current directory?

Answer Area

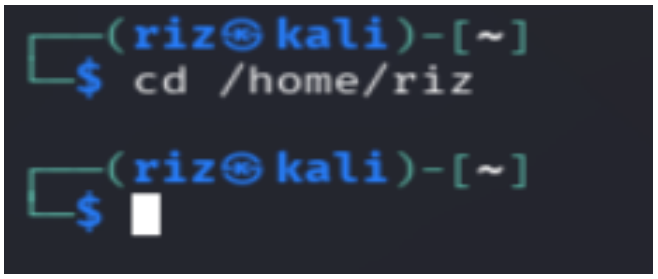
Enter your answer here

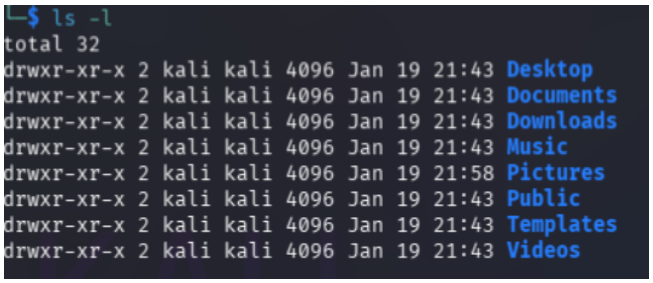
Show Answer

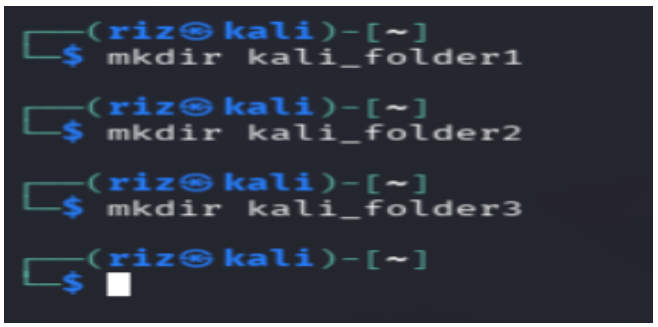
Answer :

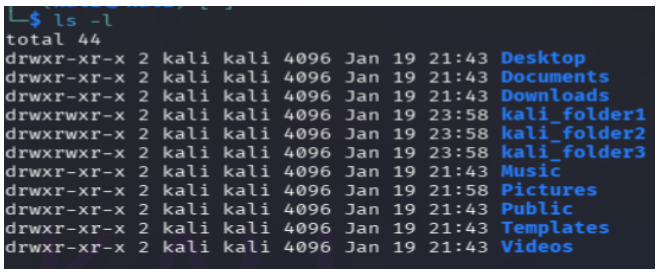


2.1









Question 2.1

```
┌──(kali㉿Kali)-[~]
└─$ cd /home/kali/kali_folder3
┌──(kali㉿Kali)-[~/kali_folder3]
└─$
```

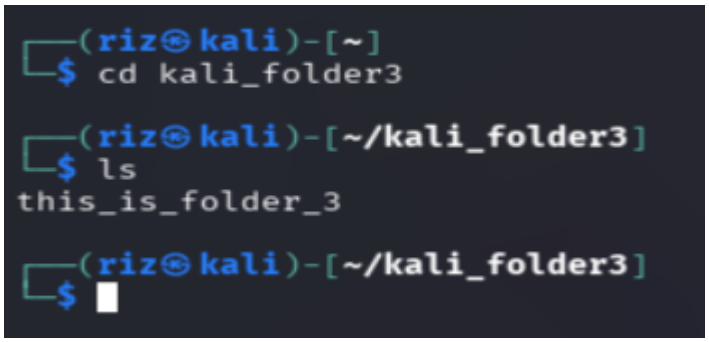
Which folder are you in now?

Answer Area

Enter your answer here

Show Answer

Answer :



2.1.1

Note: In the **[kali@kali ~]$** prompt above: The tilde symbol ~ represents the current user's home directory. In this example, the current user's home directory is **/home/kali**. After the **cd /home/kali/kali_folder3** command, the current working directory is now **/home/kali/kali_folder3**.

Note: **$** (dollar sign) indicates regular user privilege. If a '**#**' (hashtag or pound sign) is displayed at the prompt, it indicates elevated privilege (**root user**).

Note: While these symbols, conventions and main concepts remain the same, the terminal window prompt is highly customizable in Linux. Therefore, the prompt structure seen in this VM will likely differ from the prompt in other Linux installations.

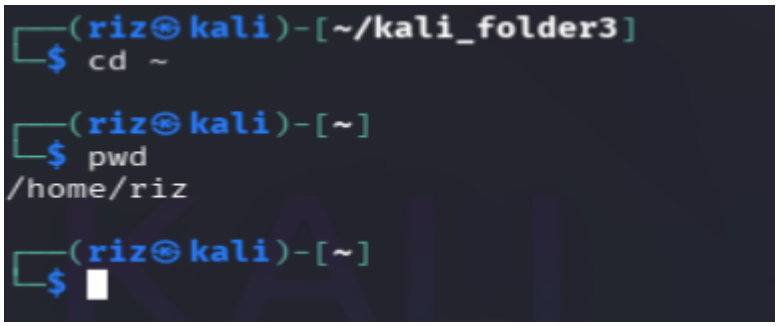
Challenge: Type the command cd ~ and describe what happens. Why did this happen?

Answer Area

Enter your answer here

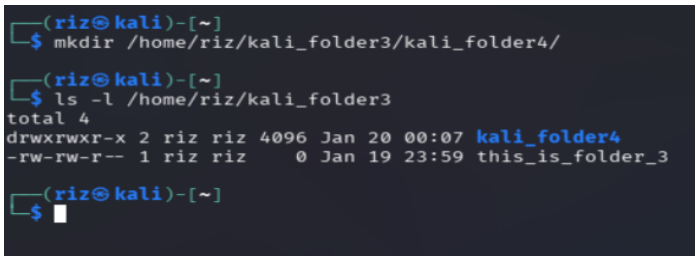
Show Answer

Answer :



Reason : When running the command "cd ~", it will change the current directory to home directory. Because the shell interprets the ~ as a shortcut for the current user's home.

2.1.2



4. Type **cd .**

```
┌──(kali㉿Kali)-[~/kali_folder3]
└─$ cd .
┌──(kali㉿Kali)-[~/kali_folder3]
└─$
```

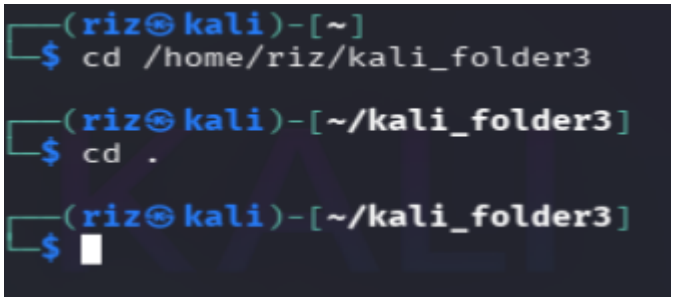
What happens?

Answer Area

Enter your answer here

Show Answer

Answer :



Nothing, the dot(.) represents the current directory.

2.1.3

5. Changing the directory to the **..** directory, will change to the directory that is one level up the path. This directory is also known as *parent directory*. Type **cd ..**

```
┌──(kali㉿Kali)-[~/kali_folder3]
└─$ cd ..
```

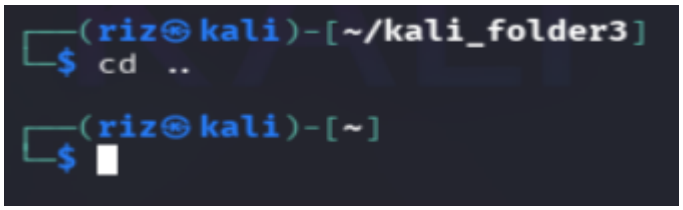
What happens?

Answer Area

Enter your answer here

Show Answer

Answer :



It will change to the previous directory, because double-dot(..) represents the previous directory.

2.1.4

If you issued the **cd ..** command at

```
┌──(kali㉿Kali)-[~]
└─$
```

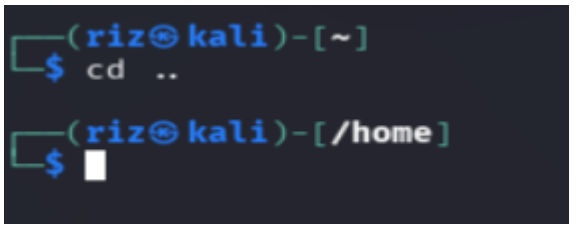
What would be the current directory?

Answer Area

Enter your answer here

Show Answer

Answer :



It will continue to change directory (cd) up one level.

2.1.5

If you issued the **cd ..** command at

```
┌──(kali㉿Kali)-[/home]
└─$
```

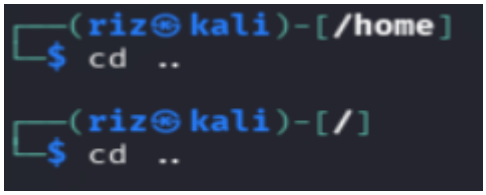
what would be the current directory?

Answer Area

Enter your answer here

Show Answer

Answer :



At the highest level of the filesystem, the root filesystem (/).

2.1.6

If you issued the **cd ..** command at

```
┌──(kali㉿Kali)-[/]
└─$
```

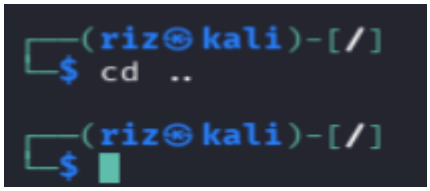
what would be the current directory?

Answer Area

Enter your answer here

Show Answer

Answer :



Using cd upon this won't change us any other directories anymore. Because we are at the highest level of the filesystem (/).

2.1.7

6. Try using the ~ to change back to the user home directory. Verify your location with the command that is used to display the current working directory.

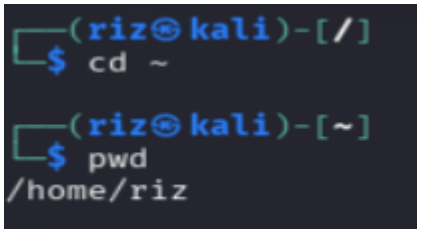
Where are you?

Answer Area

Enter your answer here

Show Answer

Answer :



3.0

b. Use the **echo** command to echo a message. Because no output was defined, echo will output to the current terminal window:

```
┌──(kali㉿Kali)-[~]
└─$ echo echo this message

echo this message
```

c. Use the **>** operator to redirect the output of echo to a text file instead of to the screen:

```
┌──(kali㉿Kali)-[~]
└─$ echo redirect this to a file > text_file.txt
```

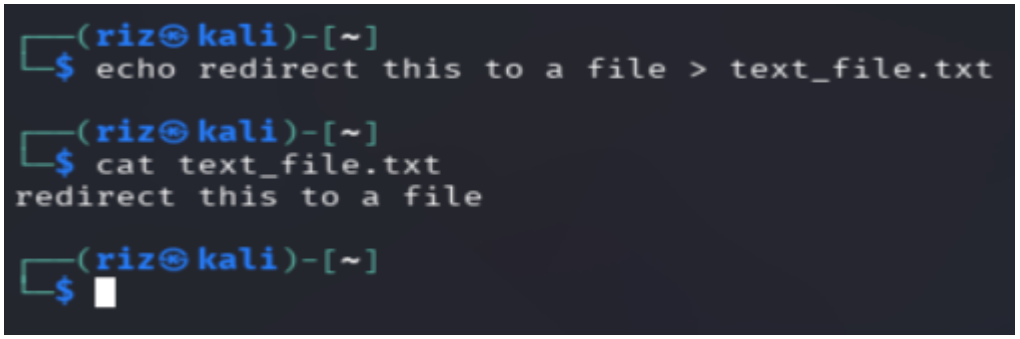
No output was shown. Is that expected?

Answer Area

Enter your answer here

Show Answer

Answer :



Yup, because the text was redirected into the file, which can be read with the “cat” command.

Step 4: Redirect and append to a text file.

a. Similar to the **>** operator, the **>>** operator also allows for redirecting data to files. The difference is that **>>** appends data to the end of the referred file, keeping the current contents intact. To append a message to the text_file.txt, issue the command below:

```
┌──(kali㉿Kali)-[~]
└─$ echo this text will be appended to the text file >> text_file.txt
```

b. Use the **cat** command to display the contents of the **text_file.txt** text file again.

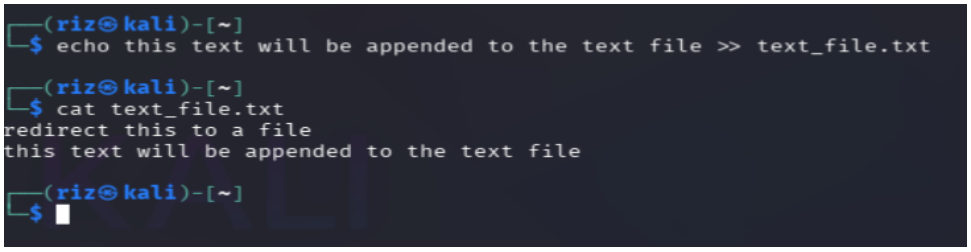
What happened to the text file? Explain.

Answer Area

Enter your answer here

Show Answer

Answer :



The new message was appended to the end of the file, keeping the original contents intact.

5.0

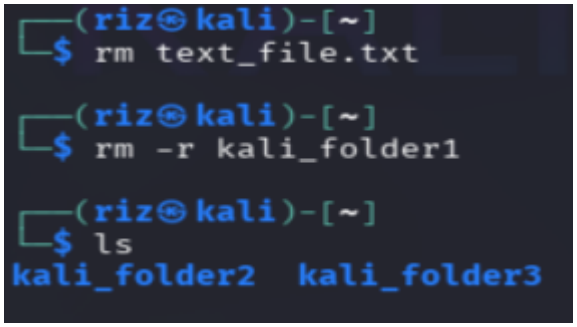
Step 5: Delete files and directories.

a. Use the **rm** command to remove files or directories. Issue the command below to remove the file **text_file.txt** from the home directory. The **ls** command can be used to show that the file **text_file.txt** has been removed from the home directory:

```
┌──(kali㉿Kali)-[~]
└─$ rm text_file.txt
```

b. In Linux, directories are seen as a type of file. Therefore, the **rm** command is also used to delete directories but the **-r** (recursive) option must be used. Notice that all files and other directories inside a given directory are also deleted when deleting a parent directory. Issue the command below to delete the **kali_folder1** folder and its contents:

```
┌──(kali㉿Kali)-[~]
└─$ rm -r kali_folder1
```



6.0

Step 6: Move files and directories.

a. Moving files works similarly to copying files. The difference is that moving a file removes it from its original location. Use the **mv** commands to move files around the local filesystem. Like the **cp** command, the **mv** command also requires source and destination parameters.
b. Create a new **text_file.txt** file in the **kali_folder2** folder by redirecting some text to it.
c. Return to the **/home/kali/** folder.
d. Issue the command below to move the **text_file.txt** from **/home/kali/kali_folder2** to the home directory. **Note**: There is a dot at the end of command.

```
┌──(kali㉿kali)-[~]
└─$ mv kali_folder2/text_file.txt .
┌──(kali㉿kali)-[~]
└─$ ls -l
total 48
drwxr-xr-x 2 kali kali 4096 Apr 10 14:51 Desktop
drwxr-xr-x 2 kali kali 4096 Apr 10 14:51 Documents
drwxr-xr-x 2 kali kali 4096 Apr 10 14:51 Downloads
drwxr-xr-x 2 kali kali 4096 Apr 21 15:25 kali_folder2
drwxr-xr-x 3 kali kali 4096 Apr 21 15:25 kali_folder3
drwxr-xr-x 2 kali kali 4096 Apr 10 14:51 Music
drwxr-xr-x 2 kali kali 4096 Apr 10 14:51 Pictures
drwxr-xr-x 2 kali kali 4096 Apr 10 14:51 Public
drwxr-xr-x 2 kali kali 4096 Apr 10 14:51 Templates
-rw-r--r-- 1 kali kali   18 Apr 21 15:49 text_file.txt
drwxr-xr-x 2 kali kali 4096 Apr 10 14:51 Videos
```

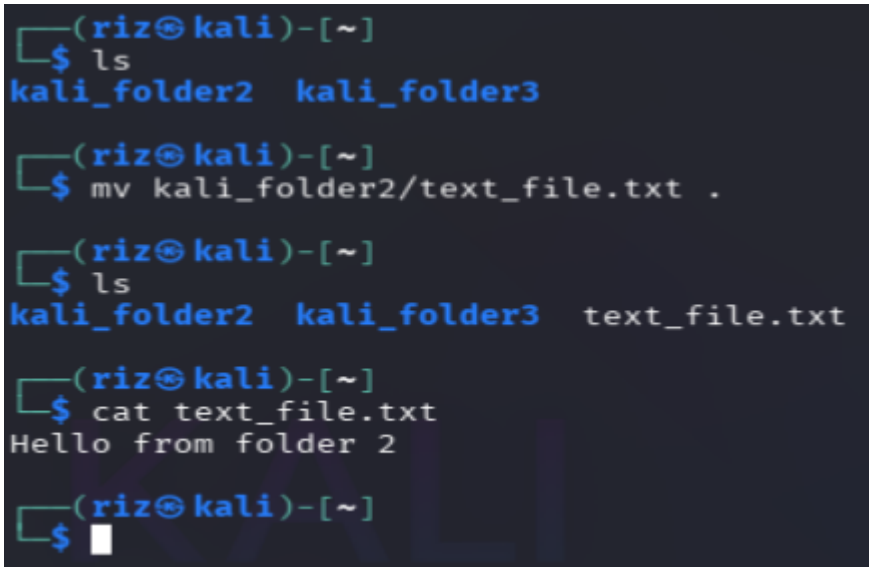
Why was the dot (".") used as the destination parameter for mv?

Answer Area

Enter your answer here

Show Answer

Answer :



Reflection

You have learned some basic ways to use Kali Linux in this lab. You learned how to use the Kali GUI and terminal with some basic Linux commands. The strength of Kali is in its collection of tools. We will cover some of those tools in labs to come.

How can you learn more about Kali command line tools?

Answer Area

You can learn more about Kali command line tools by using the man command or the --help flag to access built-in manuals and usage guides.

Show Answer